

Structural Identifiability of Epistemic Value in Active Inference

A Matched-Channel Experimental Design

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METHOD PROPOSED + SYNTHETICALLY IDENTIFIABLE + HUMAN VALIDATION PENDING

1. The identifiability problem

Active Inference (AIF) scores policies by expected free energy, which decomposes into a pragmatic term and an epistemic expected-information-gain term. Observing information-seeking choices does not identify that decomposition. Intrinsic-motivation models, Bayesian experimental design, information-directed sampling, and several human question-selection utilities can produce similar sampling. The relevant competitor class is therefore not merely reward-based reinforcement learning, but **alternative information-seeking objectives**.

We study a reduced one-step controller derived from expected free energy under stated assumptions (binary hidden state, finite observations, one-step sampling, fixed preferences, Bayesian updating). The reduced score is $V = V_{\text{instr}} + \alpha_{\text{epi}} I(Z; Y)$. This tests an EFE-derived epistemic-value component. It does not test the entirety of Active Inference. The intended ladder is general FEP $\rightarrow$ specific EFE/AIF model $\rightarrow$ identifiable prediction $\rightarrow$ strong rival. This note occupies only the last two rungs.

2. Why ordinary tasks fail (v4–v5)

Reward-only and flexible Bayes-adaptive instrumental rivals were separable when the epistemic weight was strong. A generic curiosity bonus of a different functional form, $V_{\text{curiosity}} = V_{\text{instr}} + \kappa H(Z)(2q-1)^Y$, destroyed that specificity. In the tested symmetric-cue grid the median minimum Jensen–Shannon divergence was approximately 5.5×10^{-7} (90th percentile $\approx 5.9 \times 10^{-6}$; maximum $\approx 2 \times 10^{-5}$). A search over 175,821 symmetric-cue pairs found no fully robust sign-reversal discriminator; the best pair reversed the ordering for about 71.8% of tested curiosity parameterizations. Information seeking, as such, is not mechanism-specific.

3. Proposed solution: matched asymmetric channels (v6)

We propose a matched-channel design. Sensitivity $a = P(Y=L|Z=L)$ and specificity $b = P(Y=R|Z=R)$ replace a single reliability q . Selected pairs hold constant the prior, mean cue accuracy, and instrumental sampling value, while varying exact mutual information $I(Z; Y)$. An exact-MI controller then predicts a sizable within-pair sampling contrast; the tested accuracy-based curiosity family predicts approximately none. *v6 discriminates the tested curiosity family; it does not discriminate all conceivable information-seeking algorithms*. This design class is not new: Nelson et al. (2010, E3 C1) already tied probability gain (0.25) while varying information gain ($I \approx 0.2158$ vs 0.1308 nats); only 12/22 humans preferred the richer channel. v6 is an AIF-framed analogue, not a first discriminator.

Pair	Prior p	Mean acc.	ΔMI (nats)	Pred. ΔP exact-MI	Pred. ΔP curiosity
1	0.70	0.780	0.143	0.277	0
2	0.80	0.846	0.138	0.270	2.22×10^{-16}
3	0.70	0.738	0.140	0.259	0
4	0.80	0.798	0.136	0.245	0

Table 1. Candidate matched pairs (synthetic predictions). At 240 synthetic trials, family recovery was 98.3% (exact-MI) and 96.7% (tested curiosity). These toy rates must not be used to choose participant N.

4. Interpretation ceiling (v7)

If another controller implements the same exact-MI objective and the same policy mapping, $Q_1(a|x) = Q_2(a|x)$ for every action and condition. A numerical grid check returned zero difference to floating-point precision. The equivalence result is not a mathematically surprising theorem; it delimits interpretation. A later positive human contrast can support exact-MI-sensitive valuation relative to specified alternatives. It cannot uniquely identify Active Inference or the Free Energy Principle.

5. What is needed next: a human feasibility pilot

The bottleneck is no longer toy-model separability. It is whether participants can learn asymmetric contingencies well enough for the contrast to survive. Three risks cannot be settled in simulation: (R1) failure to learn a and b ; (R2) between-subject heterogeneity that attenuates the contrast; (R3) simple heuristics that reproduce the signature. A pilot should estimate learnability, interface comprehension, empirical variance, subjective estimates of a and b , sampling-rate floor/ceiling, and candidate heuristics. It is a feasibility study, not a confirmatory test. Final N and decision thresholds will be set only after hierarchical simulation-based calibration (planned v9; not yet run).

Ask. I can contribute the design, reduced models, and adversarial comparison pipeline. I am looking for a collaborator with human behavioral-experiment infrastructure to co-develop a small feasibility pilot and, if warranted, later freeze a preregistered confirmatory protocol. The first question is not “is FEP true?” but: how do we experimentally distinguish exact information gain from merely generic information-seeking?

This side can contribute	Sought from a collaborator
Matched-channel task geometry; reduced EFE mapping; adversarial v1–v7 sequence	Judgment of human learnability and interface for asymmetric cues
Simulation code for the shipped v1 checks; preprint and this brief	Recruitment, ethics / IRB pathway, and local experimental expertise
A planned hierarchical calibration (v9) before any confirmatory N is frozen	Joint interpretation; later, if warranted, a frozen confirmatory protocol

Table 2. Proposed division of labor for a conversation, not a frozen authorship agreement.

6. What this brief does not claim

No “first” or “breakthrough” claim is made. Design-class novelty is not supported given Nelson et al. 2010. No new human data have been collected here. The current protocol is preregistration-**oriented**, not preregistration-ready. Entropies in the simulations are in nats: for $p=0.64$, $q=0.95$, $I(Z;Y)\approx 0.4625$ nats (a value near 0.953 is bits). Available now: preprint v1.1, this brief, and a one-command reproduction of the v1 mutual-information and reward-only discrimination checks. The full v4–v8 search pipeline is documented in the preprint and is not yet shipped as a public end-to-end script.

Selected references: Friston et al., Cogn Neurosci 2015; Da Costa et al., J Math Psychol 2020; Smith, Friston & Whyte, J Math Psychol 2022; Chou et al., J Math Psychol 2026; Nelson, Psychol Rev 2005; Russo & Van Roy, NeurIPS 2014.